3.1 Pairs of Lines and Angles

Geometry 3.1 · Student sheet

Name _____ Period ____ Date _____

I can…

- identify parallel lines, skew lines, and parallel planes.
- use the Parallel and Perpendicular Postulates.
- identify corresponding, alternate interior, alternate exterior, and consecutive interior angles.

Vocabulary · fill in the missing word

Parallel lines — coplanar lines that do not _____

Skew lines — lines that do not _____ and are not coplanar

Parallel planes — planes that do not _____

Transversal — a line that _____ two or more coplanar lines at different points

Part 1 · Discovery: eight angles

Draw a transversal across two lines of your notebook paper. Number the angles like the diagram on the TV and measure them.

∠1	∠2	∠3	∠4	∠5	∠6	∠7	∠8

Which pairs have the same measure?

Part 2 · Notes and examples (fill in each note, then work the example)

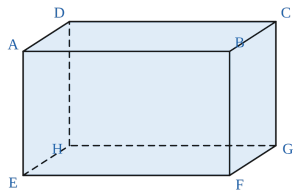
Example 1

Use the box.

a. Name a line parallel to line AB.

b. Name a line skew to line AB.

c. Name a plane parallel to plane ABCD.



Parallel Postulate: if there is a line and a point not on the line, then there is exactly _____ line through the point parallel to the given line. (The Perpendicular Postulate says the same about perpendicular lines.)

Example 2

Point P is not on line ℓ. How many lines through P are parallel to ℓ? How many are perpendicular to ℓ?

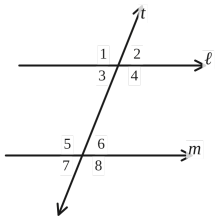
Corresponding angles are in the _____ position at each intersection. **Alternate interior angles** lie between the two lines on _____ sides of the transversal.

Example 3

Use the diagram.

a. Name all pairs of corresponding angles.

b. Name both pairs of alternate interior angles.



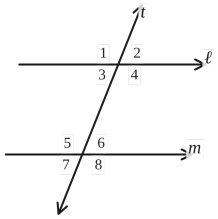
Alternate exterior angles lie _____ the two lines on opposite sides of the transversal. **Consecutive interior angles** lie between the two lines on the _____ side of the transversal.

Example 4

Use the diagram.

a. Name both pairs of alternate exterior angles.

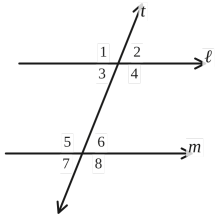
b. Name both pairs of consecutive interior angles.



Example 5

Use the diagram. Classify each pair.

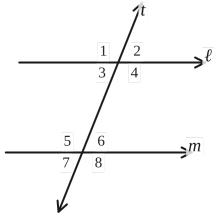
a. ∠2 and ∠7 b. ∠4 and ∠6 c. ∠1 and ∠5



Part 3 · Partner practice

Partner 1

Use the diagram. Classify each pair: ∠3 and ∠6; ∠2 and ∠6.



Partner 2

Use the diagram. Name both pairs of alternate exterior angles.